

ABSTRACT

This study addresses the critical analysis and auditing of public invoices at the Court of Accounts of the State of Rio de Janeiro (TCE/RJ), focusing on optimizing the management of government resources through machine learning technologies. It is observed that significant challenges arise from inconsistencies and the lack of standardization in the public spending invoices, which hinder effective oversight and pattern identification. This work proposes an innovative methodology that integrates advanced techniques of Natural Language Processing (NLP) and deep learning algorithms for clustering, aiming to overcome the limitations of fragmented and incomplete data. The study is based on a model capable of handling the clustering of complex and large-scale data, such as public records, providing a more robust analysis of semantic relationships between entries. The developed model enables more efficient management of public resources by identifying patterns and generating valuable insights.

Keywords: Invoices Auditing, Court of Accounts, Natural Language Processing, Deep Learning, Data Clustering, Government Resource Management, Public Data Analysis.